

Name: _____

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MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

We determined that having a functionally coded detection system that is able to trigger our notification system was going to be our highest priority going into the project. As per our initial research on fire detection systems, we evaluated that efficiently detecting and quickly alerting would be the basis of safety in our fire detection simulation. The ideal detection system we decided would be one that detects the heat from fire. Without a proper detection system with these characteristics, a notification system would not be able to perform its critical purpose of ensuring less risk of injury. Our primary stakeholder raised ideas and emphasized the importance of the detection system, which is where we would get the idea for the fire triggering a blinking light in the instance of a resident is deaf.

Design Requirement #2 (HIGH Priority)

We also easily determined that having a functionally coded notification system that is able to emit a constant, noisy, beeping was going to very closely be our second highest priority. With the evaluation of efficient detecting and quickly alerting would be critical within our fire detection system. The characteristics specifically wanted to prioritize would be fast detection and a clear and constant emittance of noise upon being triggered by the heat of fire. Proper notification prevents the risk of injury for inhabitants and gets firefighters there which could potentially reduce the property damage. Our primary stakeholder raised ideas and emphasized the importance of the detection system, which is where we would go forward with creating a fire alarm system that emitted a loud beeping.

Design Requirement #3 (HIGH Priority)

After successfully constructing a detection and notification system, a suppression system that would be able to use water to stunt the spreading of a fire. It would further the

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reduction of property damage by actively working to at least diminish the extent of a dangerous fire. Without a suppression system, the notification system and detection system would be functioning for no point. Our primary stakeholder had said our brainstormed solution for our suppression system, using water balloons, was unrealistic and could potentially take points off of our project. He also raised concerns about it even being a probable solution to fire suppression. Nonetheless, he gave us the greenlight and creative freedom to go forward with this design.

Design Requirement #4 (HIGH Priority)

Requiring the sufficient amount of materials to construct our project was a high priority requirement for our project. Having a sufficient amount of materials allows us to construct the basis of our project. With a house with a weak fixture, it is making the property more susceptible to fire damage and increasing the risk of danger to any inhabitants. Materials we needed to build included. It was supplied by our primary stakeholder and was stressed to be very important or the project could not be completed and built systems could not even be implemented.

Design Requirement #5 (HIGH Priority)

Accurate measurements as part of our project was a high priority when creating our project. This would be from the precise measurements of the walls to the exact same measurements of twin blankets. Every part of the project was stressed to be the finalized measurements, those appropriately determined by primary stakeholder design restraints and others, in which we had freedom to decide, was promptly documented. As members of this project, everyone was held accountable for double checking and documenting measurements. As mentioned, the primary stakeholder had set certain measurement constraints and, as a group, that was strictly followed.

Design Requirement #6 (LOW Priority)

Constructing furniture to furnish the interior of our house was a low priority in our project. The furniture included in our build are the chairs, beds, and desks. Under specific constraints for dimensions, that was the only thing that created any difficulty in completing it. It was easily considered one of the easier parts of the project to complete. Our primary stakeholder did not stress any importance around furniture other than, of course, getting it done.

Design Requirement #7 (LOW Priority)

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The interior design of our house was also considered a low priority in our project. This includes the rugs, curtains, covered mattresses, and pillows. Although the layout of our house was important in order to meet the criteria of being livable, It was not a priority when it came to the goal of testing to prevent a realistic house fire situation. According to our primary stakeholders priorities, the systems which would operate under the fire protection system came first. Ultimately, most if not every stakeholder had implemented this part of the design last because of the lack of urgency it held.

Rubric

Element C: Presentation and Justification of Solution Design Requirements							
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of		score = 0 No
C2. Requirements list. Design requirements _____.	score = 5 are listed, prioritized, clear, and detailed	score = 4 are listed, prioritized, generally clear, and detailed	score = 3 are listed, prioritized, generally clear, and somewhat detailed	score = 2 are listed and prioritized but some or all may be incomplete and/or lack specificity	score = 1 are listed, formatted, and prioritization is attempted, but these may be partial or overly general		score = 0 are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as	score = 5 Five	score = 4 Four	score = 3 Three	score = 2 Two	score = 1 One		score = 0 None

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high priority.	score = 5	score = 4	score = 3	score = 2	score = 1		score = 0
C4. Objective & measurable. Design requirements _____.	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable		are not measurable
C5. Leading to solution. Design requirements _____.	would be highly likely to lead to a tangible and viable problem solution	would be likely to lead to a tangible and viable problem solution	have a potential to lead to a tangible and viable problem solution	have some potential to lead to a tangible and viable problem solution	have little potential to lead to a tangible and viable problem solution		cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of		score = 0 no primary stakeholder groups.